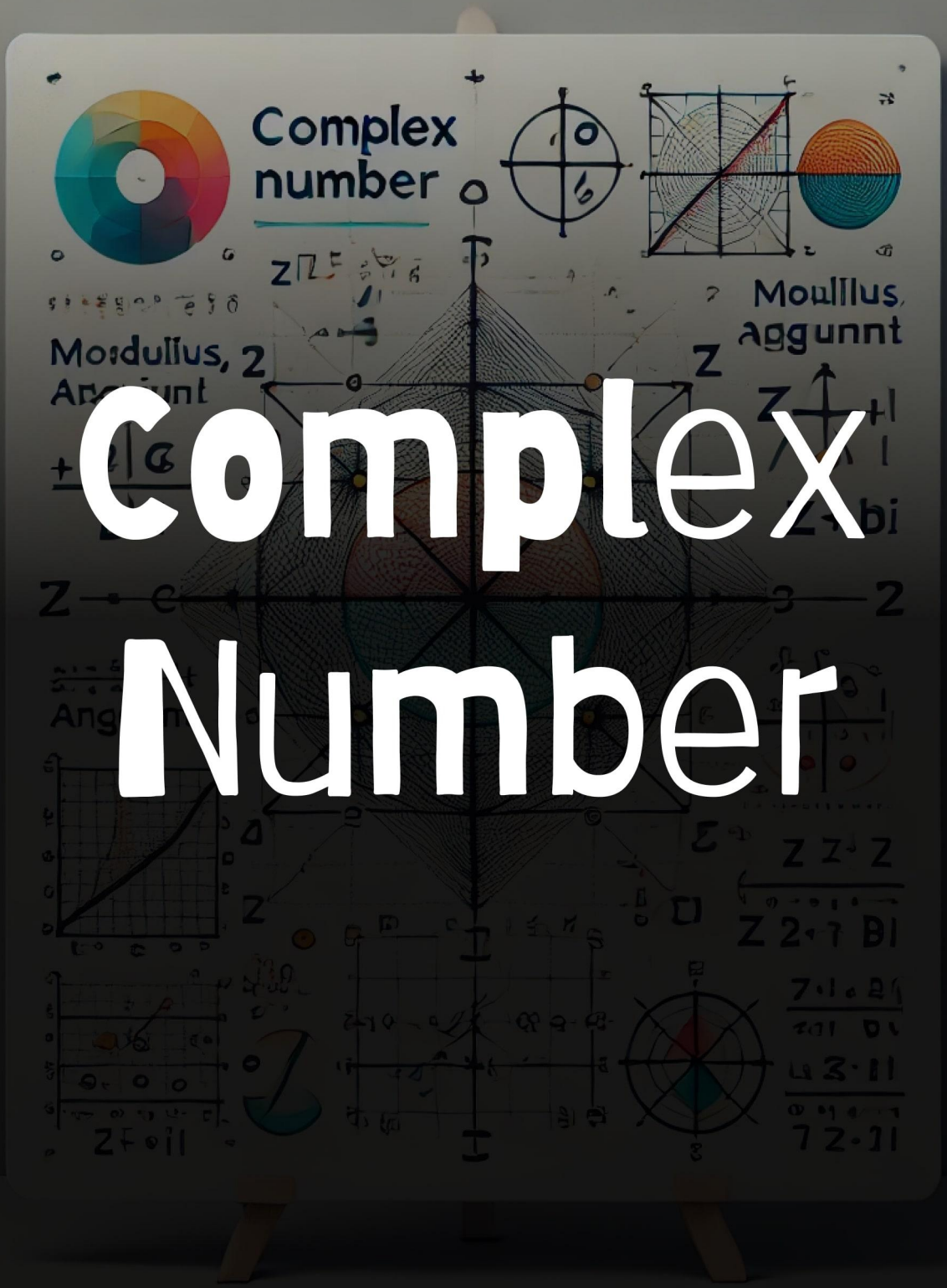

Complex
number
Modulus, 2
Modulus,
Complex
Number

Complex Numbers

$$Z = \underbrace{x}_{\text{Real part } Re(z)} + \underbrace{iy}_{\text{Imaginary part } Im(z)}$$

$i = \sqrt{-1}$
$i^2 = -1$
$i^3 = -i$
$i^4 = 1$

$Z = 2 + 3i$
$\downarrow$ $Re(z)$ $\quad$ $\downarrow$ $Im(z)$

Also, $i^{1728} = (i^4)^{432} = 1$

$i^{1727} = (i^4)^{431}(i)^3 = -i$

Ques: $z = \left(\frac{1}{4-3i}\right)$ $\quad Re(z) = ?$ $\quad Im(z) = ?$

Solu: $z = \left(\frac{1}{4-3i}\right)\left(\frac{4+3i}{4+3i}\right) = \frac{4+3i}{16+9} =$

$$\underset{\downarrow Re(z)}{\frac{4}{25}} + \underset{\downarrow Im(z)}{\frac{3}{25}i}$$

Points to Remember...

1. If $A, B \in R$ then $z = A + iB$ is a complex no., but if $z = A + iB$ is a complex no. then it is not necessary that $A, B \in R$

Example:- $z = (2+3i) + (2+4i)i$ → is a complex no. but A, B are not real

2. If imaginary part of complex no. is zero, then z is known as purely real

Example:- $z = 4$

3. If real part of complex no. is zero, then z is purely imaginary

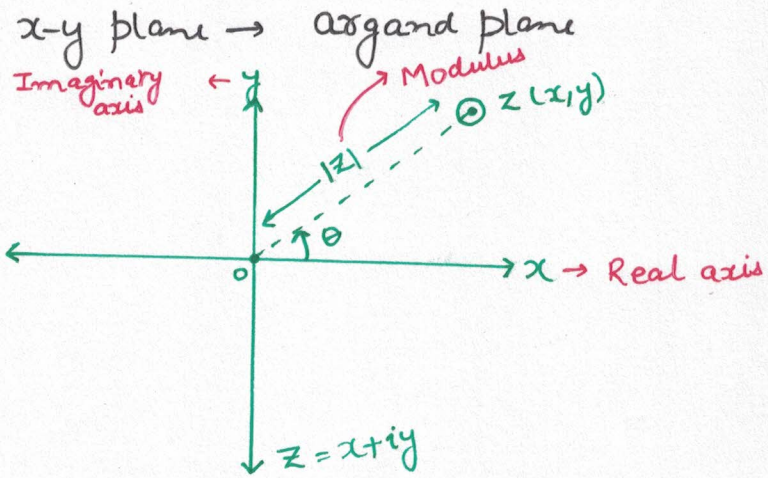
Example:- $z = 4i$

4. If $B \neq 0$, then z is imaginary number

5. A complex no. which is imaginary ($B \neq 0$) is non-comparable

6. In an argand plane, the complex no. is a point, where argand plane is nothing but a co-ordinate axes,

$x \rightarrow$ real axis $y \rightarrow$ imaginary axis



7. **Modulus of z** : The distance between the origin and point z in argand plane is called magnitude or modulus of z.

$$|z| = \sqrt{x^2+y^2}$$

OR

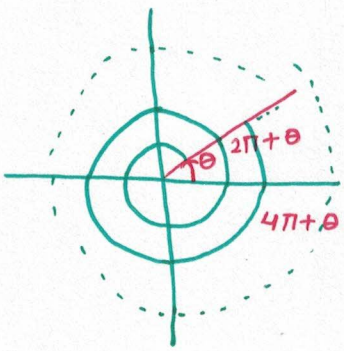
$$|z| = \sqrt{(Re(z))^2 + (Im(z))^2}$$

8. **Argument of z** : The angle made by the line segment with +ve direction of x-axis in anticlockwise is known as argument of z.

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{Im(z)}{Re(z)}\right)$$

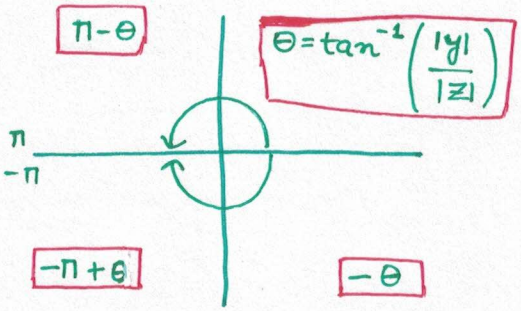
$$\arg(z) = \theta$$

General argument | **Principle argument** | **Lead +ve argument**



∞ arguments

$$\arg(z) = \theta + 2n\pi \quad \{n \in I\}$$



$$amp(z) = \theta \quad \theta \in [-\pi, \pi]$$

$$\arg(z) = \theta \quad \theta \in [0, 2\pi]$$

☞ arg (z) behaves like log while principle argument amp (z) does not behave like log

Ques: $z = -3 - 3\sqrt{3}\,i$

$\arg(z) = \tan^{-1}(\sqrt{3}) = \pi/3$

gen. $\arg(z) = \frac{4\pi}{3} + 2n\pi$

$amp(z) = -2\pi/3$

Least +ve $\arg(z) = \frac{4\pi}{3}$

Ques: Find the $\arg(z) = ?$ $\quad z = \dfrac{(1+i)^7 (\sqrt{3}+i)^9}{(1+\sqrt{3}\,i)^5}$

$\arg(z) = 7\arg(1+i) + 9\arg(\sqrt{3}+i) - 5\arg(1+\sqrt{3}i)$

$= 7(\pi/4) + 9(\pi/6) - 5(\pi/3)$

Arg (z) behaves like log

$$\arg(z^n) = n \arg(z)$$
$$\arg(z_1 z_2) = \arg z_1 + \arg z_2$$
$$\arg(z_1 / z_2) = \arg z_1 + \arg z_2$$

DIFFERENT FORMS OF COMPLEX NUMBERS

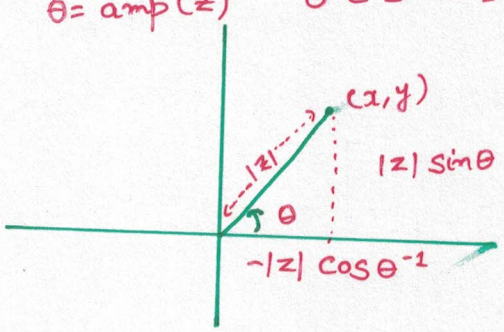
1. Co-Ordinal Form

$$z = x + iy \qquad [x, y \in R]$$

2. Polar Form

$$z = |z| \cos\theta + i|z| \sin\theta$$
$$z = r\cos\theta + ir\sin\theta$$
$$z = r(\cos\theta + i\sin\theta)$$
$$z = r(\text{cis}\,\theta)$$

$\theta = \text{amp}(z)$ $\quad \theta \in [-\pi, \pi]$



3. EULAR'S Form

$$z = r(e^{i\theta}) \qquad e^{i\theta} = \text{cis}\,\theta$$

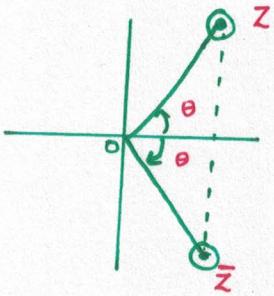
eg: $$z^n = r^n (e^{i\theta})^n = r^n (e^{i\theta n})$$
$$\arg(z^n) = n \arg(z)$$

eg: $$z_1 z_2 = r_1 r_2 \, e^{i\theta_1} \cdot e^{i\theta_2}$$
$$= r_1 r_2 \, e^{i\theta_1 + i\theta_2}$$

$$\arg(z_1 z_2) = \theta_1 + \theta_2$$
$$= \arg(z_1) + \arg(z_2)$$

4 Conjugate of z

If $z = x + iy$, then $\bar{z} = x - iy$



NOTES

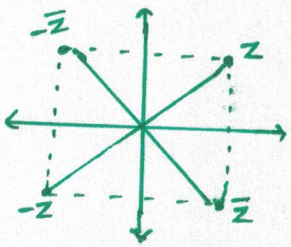
1. $z + \bar{z} = 2\,Re(z)$

2. $z - \bar{z} = 2i\,Im(z)$

3. $\bar{z} = z \Rightarrow$ z is purely real

4. $\bar{z} = -z \Rightarrow$ z is purely imaginary

5. $z \cdot \bar{z} = |z|^2 = x^2 + y^2$



6. $z, \bar{z}, -z, -\bar{z}$ forms a rectangle

7. Two complex no.'s are said to be equal if their corresponding real parts as well as imaginary parts are equal.

   i.e. z_1 and z_2 are said to be equal if,

   $|z_1| = |z_2|$ as well as $\arg(z_1) = \arg(z_2)$

8. If $z_1 = x_1 + iy_1$ $\quad z_2 = x_2 + iy_2$, then

$$z_1 + z_2 = x_1 + x_2 + i(y_1 + y_2)$$

$$z_1 - z_2 = x_1 - x_2 + i(y_1 - y_2)$$

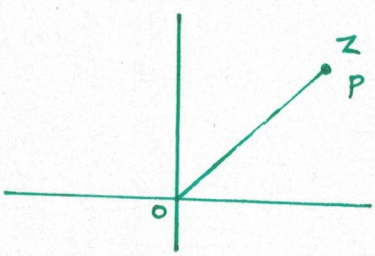
$$z_1 \cdot z_2 = x_1x_2 - y_1y_2 + i(x_1y_2 + x_2y_1)$$

$$\frac{z_1}{z_2} = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i\,\frac{(x_2y_1) - x_1y_2}{x_2^2 + y_2^2}$$

$$\frac{z_1}{z_2} \times \frac{\bar{z}_2}{\bar{z}_2} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{x_2^2 + y_2^2}$$

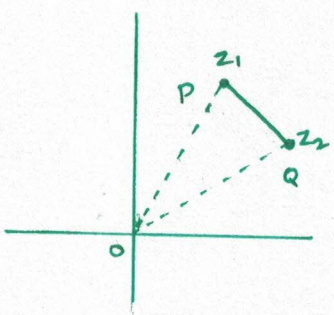
$$= \frac{x_1 x_2 + y_1 y_2 + i(x_2 y_1 - x_1 y_2)}{x_2^2 + y_2^2}$$

9. Complex no. in argand plane can be treated as vector.



$\vec{OP} = z$

$|z| = |\vec{OP}|$



$\vec{PQ} = z_2 - z_1$

$|PQ| = |z_1 - z_2|$

Here, $|z_1 - z_2|$ represents the length of line segment joining

$$|\vec{PQ}| = |z_2 - z_1| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Ques: Find the locus of z, which satisfies the relation: $|z-2| = |z-4|$

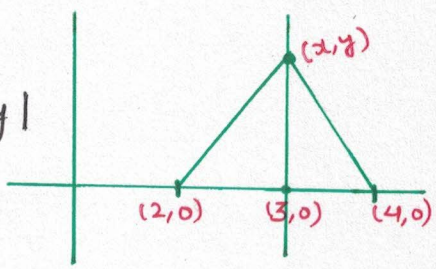
Solution: if $z = x + iy$

$$|(x-2) + iy| = |(x-4) + iy|$$

$$(x-2)^2 + y^2 = (x-4)^2 + y^2$$

$$x - 2 = x - 4$$

$$2x = 6 \Rightarrow \boxed{x = 3}$$



NOTE $|z - z_1| = |z - z_2| \Rightarrow$ { z is the perpendicular bisector joining the line segment of z_1, z_2

Ques: Find the locus of z, which satisfy $|z-(2+3i)|=4$

Solution:

$$|(x-2)+i(y-3)| = 4$$

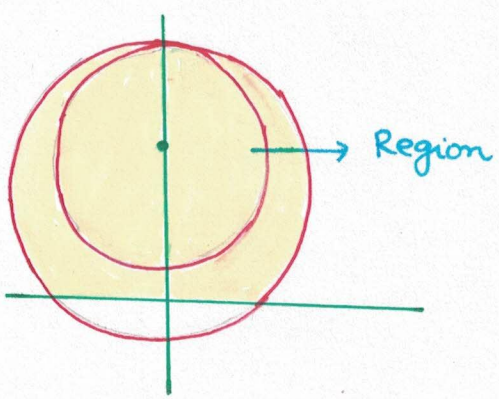
$$(x-2)^2 + (y-3)^2 = 16 \rightarrow \boxed{\text{Circle}}$$

NOTE

$|z-z_1| = a$ } z is the locus of circle with centre z_1 and radius a

Ques: Find the region satisfying $|z-2i|<3$ and $Im(z)\geq 0$

Solution:



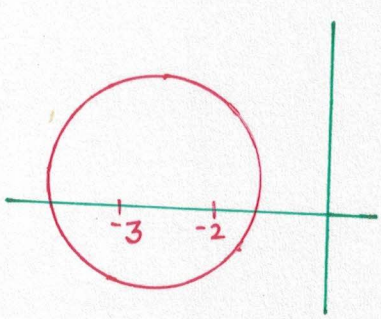
$$x^2 + (y-2)^2 < 3$$

Ques: $|z+3-i| \geq -2$, find z

Solution:

$\sqrt{(x+3)^2 + (y-1)^2} \geq -2$ always true

Whole argand plane



Ques: Find the locus of z, if $\arg(z) = \pi/4$

Solution:

$$z = x + iy$$

$$1 = y/x$$

$$\Rightarrow \quad x = y \rightarrow \boxed{\text{Straight Line}}$$

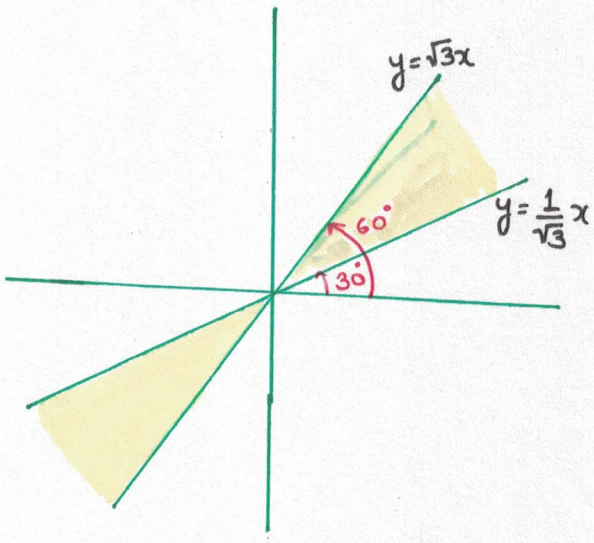
Ques: a) Find the area of the region bounded by-

$$\frac{\pi}{6} \leq \arg(z) \leq \frac{\pi}{3}$$

$z = x + iy$

$\frac{1}{\sqrt{3}} x \leq y$

$\frac{1}{\sqrt{3}} \leq \frac{y}{x} \leq \sqrt{3}$



$$\text{area} = 2\int \sqrt{3}x\,dx - \int \frac{1}{\sqrt{3}} x\,dx$$

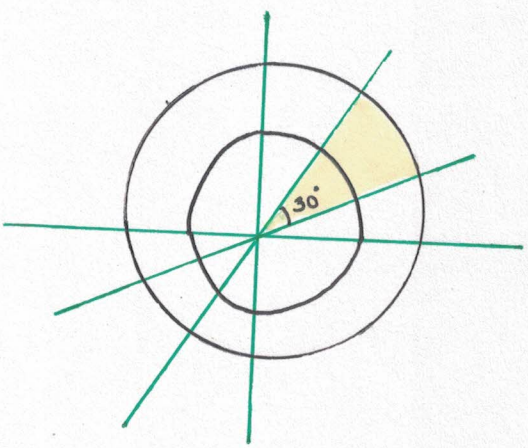
$$= \int_0^{\infty} \sqrt{3}\,\frac{x^2}{2} - \frac{1}{\sqrt{3}}\,\frac{x^2}{2}$$

B) Find the area $1 < |z| \leq 3$

$$\text{area} = \frac{1}{2} \times r\theta$$

$$= \frac{1}{2} \times \frac{\pi}{6} \times 4 \times 2$$

$$= \frac{2\pi}{3}$$



Ques: Find the complex no. having least +ve argument and satisfying the inequality $|z-3+3i| \leq 3$

solution

$$\sqrt{(x-3) + i(y+3)} \leq 3$$

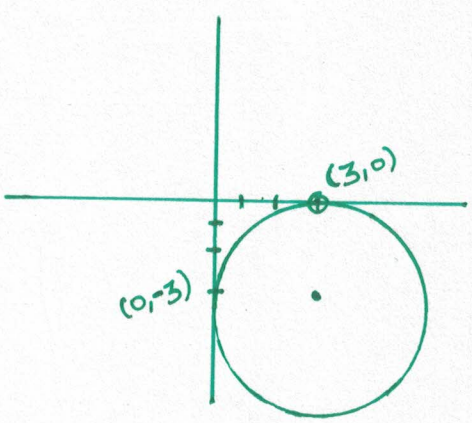
$$(x-3)^2 + (y+3)^2 \leq 9$$

$$(x-3)^2 + 9 = 9$$

$$x = 3 \Rightarrow \boxed{z = 3}$$

$z = x + iy$

$\theta = \tan^{-1}\frac{y}{x}$



PROPERTIES OF $\bar{z}$

- $\overline{z_1 \pm z_2} = \bar{z}_1 + \bar{z}_2$
- $\overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$
- $\overline{\frac{z_1}{z_2}} = \frac{\bar{z}_1}{\bar{z}_2}$
- $\overline{z^n} = (\bar{z})^n$

PROPERTIES OF $|z|$

- $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$
- $|z^n| = |z|^n$
- $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$
- $|z| = 1 \Rightarrow z\bar{z} = 1 \Rightarrow \bar{z} = \frac{1}{z}$
- $|z_1 + z_2| \leq |z_1| + |z_2|$
- $||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$ ⟵ Triangle Inequality

QUESTIONS

Question $|z - i| = 1$ $\quad \arg(z) = \frac{\pi}{2}$ $\quad z = ?$

Solution

$|z| = x + i(y-1)$

$z = (y-1)i$

$z = 2i$

$\frac{y}{x} = \tan\left(\frac{\pi}{2}\right)$ $\quad x \neq 0$

$y - 1 = 1$

$y = 2$

Question $|z + 3 - i| = 4$ $\quad \arg(z) = \pi/4$

Solution

$(x+3)^2 + (y-1)^2 = 16$

$(y-1)^2 = 7$

$y = \sqrt{7} + 1$

$(x+3)^2 + (x-1)^2 = 16$

$2x^2 + 9 + 1 + 6x - 2x = 16$

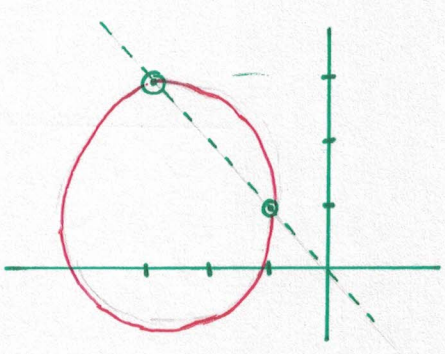
$2x^2 + 4x - 6 = 0$

$x^2 + 2x - 3 = 0$

$x(x+3) - 1(x+3) = 0$

$x = 0 \quad x = 1$

$x = -3$



Ques: Solve for z :

a) $z^2 - iz - 1 = 0$

$$\Rightarrow \quad z = \frac{i \pm \sqrt{i^2+4}}{2} = \frac{i \pm \sqrt{3}}{2}$$

$$\Rightarrow \quad z\bar{z} - i\bar{z} - 1 = 0$$

$$\Rightarrow \quad \bar{z}(z - i) - 1 = 0$$

$z = x + iy$
$\bar{z} = x - iy$

$$\Rightarrow \quad \bar{z}(z - i) = 1$$

$$\Rightarrow \quad (x - iy)[x + i(y-1)] = 1$$

$$\Rightarrow \quad x^2 + y(y-1) + i$$

$$\Rightarrow \quad (x+iy)(x - iy) - i(z) = 1$$

$$\Rightarrow \quad x^2 - y^2 + i(xy + xy) - i(z) = 1$$

$$\Rightarrow \quad x^2 - y^2 - 1 = i(z - 2xy)$$

$$\Rightarrow \quad x^2 - y^2 + i\,2xy = 1 + i(z)$$

$$\Rightarrow \quad 2xy = 2$$

$$xy = 1$$

$x^2 - y^2 = 1$
$x^2 = 1 + y^2$

$$\Rightarrow \quad [\sqrt{1+y^2}]\, y = 1$$

$$\Rightarrow \quad (1+y^2)\, y^2 = 1$$

$y^2 = t$

$$\Rightarrow \quad t + t^2 = 1$$

$$\Rightarrow \quad t^2 + t - 1 = 0 \quad \Rightarrow \quad t = \frac{-1 \pm \sqrt{5}}{2}$$

$$\Rightarrow (x+iy)^2 - (x - iy)i - 1 = 0$$

$$\Rightarrow (x^2 - y^2 + 2ixy) - (x - iy)i - 1 = 0$$

$$\Rightarrow (x^2 - y^2 - y - 1) + i(2xy - z) = 0 + i0$$

$$\Rightarrow \quad x^2 - y^2 - y - 1 = 0$$

$$\Rightarrow \quad x = \pm \frac{\sqrt{7}}{2}$$

$2xy - x = 0$
$x(2y - 1) = 0$
$y = 1/2, \; x = 0$

b) $iz^3 + z^2 - z + i = 0$

$\Rightarrow z(iz^2 + z - 1) + i = 0$

$\Rightarrow i(x^2 - y^2 + 2ixy)(x + iy) = 0$

$\Rightarrow$ Sum $= \frac{-1}{i}$ $\quad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{-1}{i}$ $\quad \alpha\beta\gamma = \frac{-i}{i} = -1$

$\Downarrow$

$\alpha\beta = \frac{-1}{\gamma}$

$\Rightarrow \alpha + \beta + \gamma = \frac{-1}{i}$

$\Rightarrow \gamma = \left(\frac{-1}{i}\right) - (\alpha + \beta)$

$\gamma(\alpha + \beta) = \frac{-1}{i} + \frac{1}{\gamma}$

$\left[\frac{-1}{i} - (\alpha + \beta)\right](\alpha + \beta) = \frac{-1}{i} + \frac{1}{\gamma}$

$z = i$ is a root

$-\frac{(\alpha + \beta)}{i} - (\alpha + \beta)^2 = 0$

$\Rightarrow i^4 + i^2 - i + i = 0$

$(z - i)(iz^2 - 1) = 0$

$\Rightarrow$

$z - i \;\big)\; iz^3 + z^2 - z + i \;\big(\; iz^2 - 1$

$-\,(iz^3 + z^2)$

$-z + 1$

$z^2 = \frac{1}{i} \Rightarrow z = \pm\sqrt{-i}$

$z = \frac{1 \pm i}{\sqrt{2}}$

$\Rightarrow z = x + iy$

$\Rightarrow x^2 + y^2 + 2xyi = -i$

$\Rightarrow x^2 = y^2 \qquad xy = -1/2$

$\Rightarrow x = -y \qquad x = \frac{1}{\sqrt{2}}$

c) Find $|z|$ and amp(z) if $z = 1 + \cos\left(\frac{6\pi}{5}\right) + i\sin\left(\frac{6\pi}{5}\right)$

$|z| = \sqrt{\left(1 + \cos\frac{6\pi}{5}\right)^2 + \left(\sin\frac{6\pi}{5}\right)^2}$

$= \sqrt{2}\sqrt{\left(1 + \cos\frac{6\pi}{5}\right)}$

$= 2\left|\cos\frac{3\pi}{5}\right|$

$= \frac{3\pi}{5}$

$\text{amp}(z) = \frac{2\sin\left(\frac{3\pi}{5}\right)\cos\left(\frac{3\pi}{5}\right)}{2\cos^2\frac{3\pi}{5}}$

$\Rightarrow \theta = \frac{3\pi}{5}$

Ques:- Find the magnitude and arg (z) ⇒ $\left[\frac{(8+i)(1+i)}{5-i}\right]^6$

$$\arg(z) = 6\arg(8+i) + 6\arg(1+i) - 6\arg(5-i)$$

$$= 6\tan^{-1}(1/8) + 6\tan^{-1}(1) + 6\tan^{-1}(1/5)$$

$$= 6\left[\tan^{-1}\frac{\frac{1}{8}+\frac{1}{5}}{1-\frac{1}{40}}\right] + 6\frac{\pi}{4}$$

$$= 6\tan^{-1}\frac{1/3}{39}\ \frac{1}{3}$$

$$= 6\tan^{-1}\frac{\left(\frac{1}{3}+1\right)}{\left(1-\frac{1}{3}\right)}$$

$$= 6\tan^{-1}(4/2) \Rightarrow \boxed{\arg(z) = 6\tan^{-1}(2)}$$

$$|z| = \left[\frac{8-1+i(8+1)}{5-i}\right]^6$$

$$= \left[\left(\frac{7+9i}{25+1}\right)(5+i)\right]^6 = \left[\frac{35+9+i(7+45)}{26}\right]^6$$

$$= \left[1+\frac{i(52)}{26}\right]^6 = (\sqrt{5})^6 \Rightarrow \boxed{|z| = 125}$$

Ques:- Let $z_1, z_2, \ldots z_n$ are complex no. in such a way that $z_1, z_2, \ldots z_n$ lying on a circle of center (0,0) and radius = 1.

If $W = \sum_{k=1}^{n}(z_k)\cdot\sum_{k=1}^{n}\frac{1}{z_k}$. Then prove that ⇒

W is real and $W \in (0, n^2)$

Hint

$|z_1| = |z_2| = \ldots |z_n| = 1$ & $\bar{z}_1 = 1/z_1$

then, $W = (z_1+z_2+\ldots z_n)\cdot\left(\frac{1}{z_1}+\frac{1}{z_2}+\ldots\frac{1}{z_n}\right)$

$$= (z_1+z_2+\ldots z_n)\cdot(\bar{z}_1+\bar{z}_2+\bar{z}_3+\ldots+\bar{z}_n)$$

$= (z_1 + z_2 + \dots z_n)\,(\overline{z_1 + z_2 + \dots z_n})$

Ques: If $|z| \le 1$ and $|w| \le 1$ then prove that :-

$$|z-w|^2 = (|z| - |w|)^2 + \{\arg(z) - \arg(w)\}^2$$

DO WELL

Ques: $\left|z - \frac{4}{z}\right| = 2$, Find max & min value of $|z|$

Solution:

$$\left|\,|z| - \left|\frac{-4}{z}\right|\,\right| \le \left|z + \left(\frac{-4}{z}\right)\right| \le |z| + \left|\frac{-4}{z}\right|$$

$$\left|\,|z| - \frac{4}{|z|}\,\right| \le 2 \le |z| + \frac{4}{|z|}$$

① $-2 \le |z| - \frac{4}{|z|} \le 2$

$t^2 + 2t - 4 \ge 0$

$$t = \frac{-2 \pm \sqrt{4+16}}{2} = \frac{-2 \pm 2\sqrt{5}}{2} = -1 \pm \sqrt{5}$$

$$= (-\infty,\ -1-\sqrt{5}) \cup (-1+\sqrt{5},\ \infty)$$

$t^2 - 4 - 2t \le 0$

$$t = \frac{2 \pm \sqrt{4+16}}{2} = 1 \pm \sqrt{5}$$

$$= [1-\sqrt{5},\ 1+\sqrt{5}]$$

$t \in [\sqrt{5}-1,\ \sqrt{5}+1]$

② $t + \frac{4}{t} \ge 2$

$|z| \in [1, \sqrt{5}, 1+\sqrt{5}]$

$t_{max} = 1+\sqrt{5}$

$t_{min} = \sqrt{5}-1$

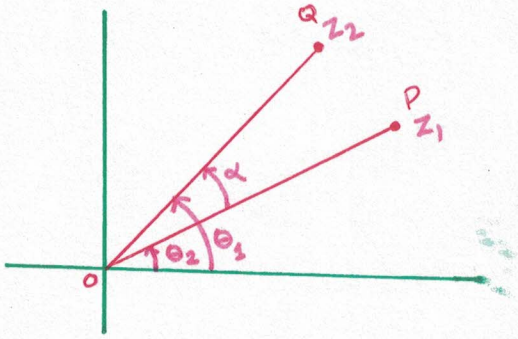
AM, GM

$$\frac{t + \frac{4}{t}}{2} \ge \sqrt{4}$$

$t + \frac{4}{t} \ge 4$

$\Rightarrow t > 0$

ROTATION OF COMPLEX NO'S



$\alpha = \theta_2 - \theta_1$

$|OP| = |OQ|$

$|z_1| = |z_2| = r$

$z_1 = re^{i\theta_1}$

$z_2 = re^{i\theta_2}$

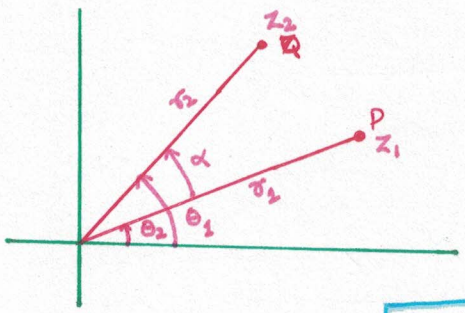
$$\alpha = \theta_2 - \theta_1 = \arg(z_2) - \arg(z_1) = \arg\left(\frac{z_2}{z_1}\right)$$

$$z_2 = re^{i\theta_2} = re^{i(\alpha+\theta_1)} = re^{i\alpha}\cdot e^{i\theta_1}$$

$$\boxed{z_2 = z_1 \cdot e^{i\alpha}}$$

where $e^{i\alpha} = \text{cis}\,\alpha$, $\alpha = \arg(z_2/z_1)$

ROTATION OF COMPLEX NUMBER



$z_2 = r_2 e^{i(\theta_1+\alpha)}$

$z_2 = r_2 e^{i\theta_1}\cdot e^{i\alpha}$

$z_2 = |z_2|\cdot\frac{|z_1|}{|z_1|}\, e^{i\theta_1}\cdot e^{i\alpha}$

$$\boxed{z_2 = \frac{|z_2|}{|z_1|}\cdot z_1 \cdot e^{i\alpha}}$$

① $|z_1 + z_2| \le |z_1| + |z_2|$

$|z_1 + z_2 + \dots z_n| \le |z_1| + |z_2| + \dots |z_n|$

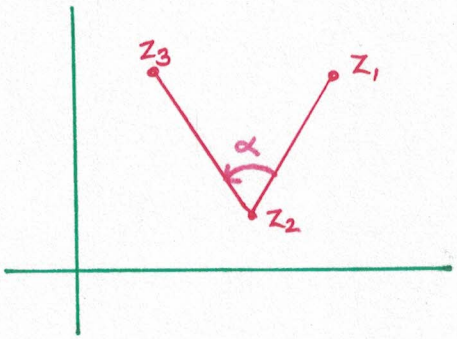
$0 \le |z_1 + z_2 + \dots z_n| \le n$

$0 \le \sqrt{w} \le n$

$0 \le w \le n^2$

② $z_1 = r_1\,\text{cis}\,\theta_1$ $\qquad z_2 = r_2\,\text{cis}\,\theta_2$

ROTATION OF A COMPLEX NO. Z_1 W.R.T. Z_2



$$|Z_1 - Z_2| = |Z_3 - Z_1|$$

$$Z_{final} = Z_{initial}\, e^{i\alpha}$$

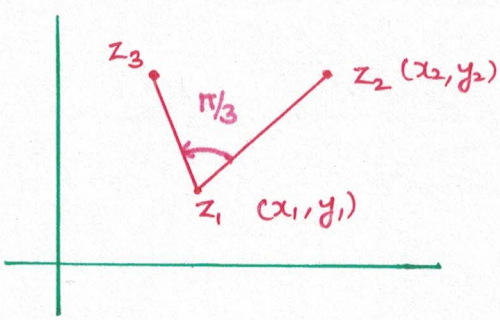
$$Z_3 = Z_2 + (Z_1 - Z_2)e^{i\alpha}$$

$$\boxed{\text{final} = (\text{initial})(e^{i\alpha})}$$

ROTATION OF A COMPLEX NO. Z_1 W.R.T. Z_2 IF MAGNITUDE IS CHANGED

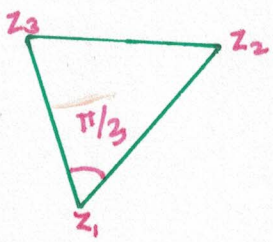
$$\boxed{Z_3 - Z_2 = \frac{|Z_3 - Z_2|}{|Z_1 - Z_2|}(Z_1 - Z_2)\, e^{i\alpha}}$$

TO FIND THE THIRD VERTEX OF AN EQUILATERAL Δ



$$\boxed{\begin{aligned} Z_3 - Z_1 &= (Z_2 - Z_1)\, e^{i\pi/3} \\ Z_3 &= Z_1 + (Z_2 - Z_1)\, e^{i\pi/3} \end{aligned}}$$

Ques: If Z_1, Z_2, Z_3 are 3 verticles of an equilateral Δ, then prove that $Z_1^2 + Z_2^2 + Z_3^2 = Z_1Z_2 + Z_2Z_3 + Z_1Z_3$



Solution:

$$Z_2 - Z_3 = (Z_1 - Z_3)e^{i\theta}$$

$$\Rightarrow \frac{Z_2 - Z_3}{Z_1 - Z_3} = e^{i\pi/3}$$

Also, $\dfrac{Z_1 - Z_2}{Z_3 - Z_2} = e^{i\pi/3}$

$$\Rightarrow \frac{Z_2 - Z_3}{Z_1 - Z_3} = \frac{Z_1 - Z_2}{Z_3 - Z_2}$$

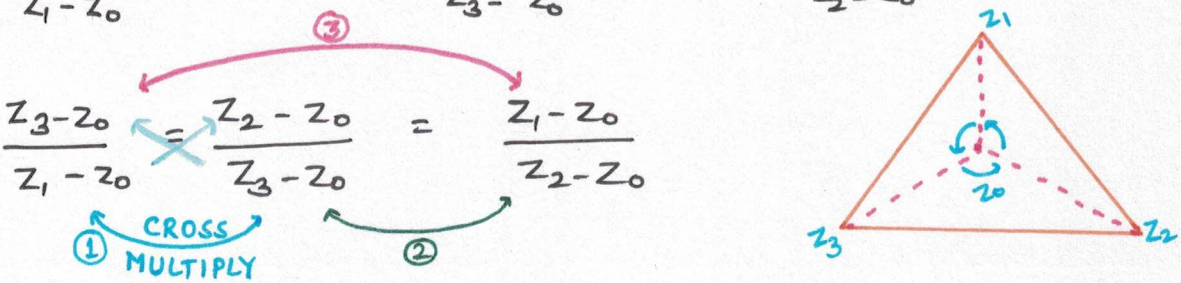
$$\Rightarrow Z_2Z_3 - Z_2^2 - Z_3^2 + Z_2Z_3 = Z_1^2 - Z_1Z_2 - Z_1Z_3 + Z_2Z_3$$

$$\Rightarrow \boxed{Z_1^2 + Z_2^2 + Z_3^2 = Z_1Z_2 + Z_2Z_3 + Z_1Z_3}$$

HENCE PROVED...

Ques: If z_1, z_2, z_3 are the vertex of an eq. Δ, and z tobe centroid, then prove that:- $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$

solu:- $\frac{z_3 - z_0}{z_1 - z_0} = e^{i\theta}$, $\frac{z_2 - z_0}{z_3 - z_0} = e^{i\theta}$, $\frac{z_1 - z_0}{z_2 - z_0} = e^{i\theta}$



$\Rightarrow z_3^2 - z_3z_0 - z_3z_0 + z_0^2 = z_1z_2 - z_1z_0 - z_2z_0 + z_0^2$

$\Rightarrow z_3^2 - 2z_3z_0 = z_1z_2 - 2z_1z_0$ (1)

Similarly $z_2^2 - 2z_2z_0 = z_2z_3 - 2z_2z_0$ (2)

and, $z_1^2 - 2z_1z_0 = z_3z_1 - 2z_3z_0$ (3)

$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$ ✓ EQUATION (A)

If $z_0 = \frac{z_1 + z_2 + z_3}{3} \Rightarrow 3z_0 = z_1 + z_2 + z_3$

$\Rightarrow 9z_0^2 = z_1^2 + z_2^2 + z_3^2 + 2z_1z_2 + 2z_2z_3 + 2z_1z_3$

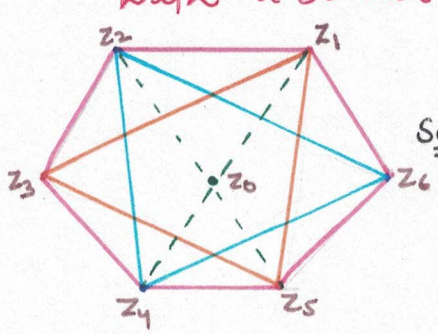
$\Rightarrow 9z_0^2 = z_1^2 + z_2^2 + z_3^2 + 2(z_1z_2 + z_2z_3 + z_3z_1)$

Refer Equation (A)

$\Rightarrow 9z_0^2 = 3(z_1^2 + z_2^2 + z_3^2)$

$\Rightarrow$ **$z_1^2 + z_2^2 + z_3^2 = 3z_0^2$**

Ques: If $z_1, z_2, z_3 \dots z_6$ are vertex of a regular hexagon with a center z_0. Prove that -



$z_1^2 + z_2^2 + z_3^2 + z_4^2 + z_5^2 + z_6^2 = 6z_0^2$

So 2 Eq. Δs : $z_1^2 + z_3^2 + z_5^2 = 3z_0^2$

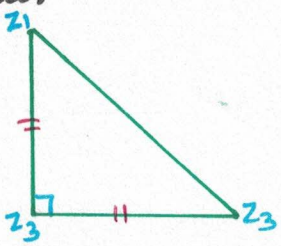
$+ \quad z_2^2 + z_4^2 + z_6^2 = 3z_0^2$

$z_1^2 + z_2^2 + z_3^2 + z_4^2 + z_5^2 + z_6^2 = 6z_0^2$

Ques: If z_1, z_2, z_3 are vertex of a right angle isosceles Δ where z_2 has right angle. Prove that:

$$z_1^2 + 2z_2^2 + z_3^2 = 2z_2(z_1 + z_3)$$

Solu:



$$|z_3 - z_2| = |z_1 - z_2|$$

$$\Rightarrow z_1 - z_2 = (z_3 - z_2)(\cos 90^\circ + i \sin 90^\circ)$$

$$\Rightarrow z_1 - z_2 = (z_3 - z_2)(i)$$

$$\Rightarrow z_1^2 + z_2^2 + 2z_1 z_2 = -(z_3^2 + z_2^2 - 2z_3 z_2)$$

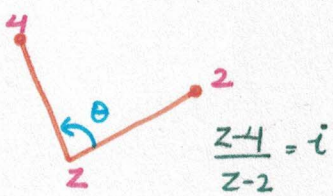
$$\Rightarrow \boxed{z_1^2 + z_3^2 + 2z_2^2 = 2z_2(z_1 + z_3)}$$

Ques: Find the locus of z if,

(a) $\arg\left(\frac{z-4}{z-2}\right) = \frac{\pi}{2}$ (b) $\arg\left(\frac{z-4}{z-2}\right) = \frac{\pi}{4}$

Solu:

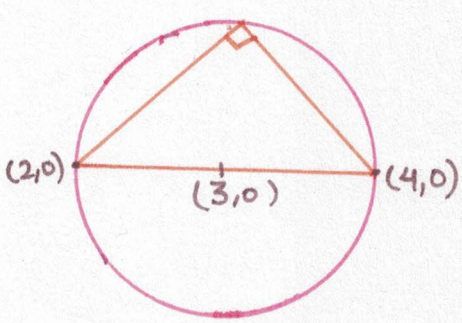
(a) $\arg\left(\frac{(x-4)+iy}{(x-2)+iy}\right) = \frac{\pi}{2}$



$$\Rightarrow \arg(z-4) - \arg(z-2) = \pi/2$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{x-4}\right) - \tan^{-1}\left(\frac{y}{x-2}\right) = \pi/2$$

$$\Rightarrow \left(\frac{\frac{y}{x-4} - \frac{y}{x-2}}{1 + \frac{y^2}{(x-4)(x-2)}}\right)$$



$$\Rightarrow (x-4)(x-2) + y^2 = 0$$

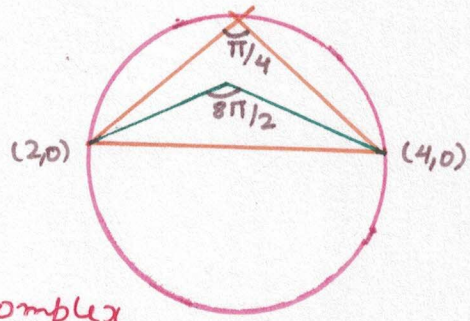
$$\Rightarrow \boxed{x^2 + y^2 - 6x + 8 = 0}$$

(b) $\tan^{-1}\left(\frac{y}{x-4}\right) - \tan^{-1}\left(\frac{y}{x-2}\right) = \frac{\pi}{4}$

$$\tan^{-1}\left(\frac{y(x-2) - y(x-4)}{(x-4)(x-2) + y^2}\right) = \tan^{-1}(1)$$

$xy - 2y - xy + 4y = x^2 + y^2 - 6x + 8$

$x^2 + y^2 - 6x - 2y + 8 = 0$



Ques: $z^2 + pz + q = 0$ is satisfied by 2 complex no.'s and angle z_1 & z_2 is α and $|z_1| = |z_2|$, then prove that - $p^2 = 4q\cos^2 \alpha/2$

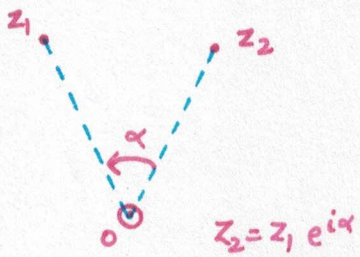
Solu: $-p = z_1 + z_2 \qquad z_1 z_2 = q$

$(z_1 - z_2)^2 = (z_1 + z_2)^2 - 4z_1 z_2$

$= p^2 - 4q$

$4q + (z_1 - z_1 e^{i\alpha})^2 = p^2$

$4q + z_1^2 (1 - e^{i\alpha}) = p^2$



Also, $2z = -p \pm \sqrt{p^2 - 4q}$

$2z_1 = -p + \sqrt{p^2 - 4q}$

$2z_2 = -p - \sqrt{p^2 - 4q}$

DE MOIVRE'S THEOREM

$\text{Cis}\theta = (\cos\theta + i\sin\theta)^n \rightarrow \cos(n\theta) + i\sin(n\theta)$

$n \in Z \rightarrow$ only 1 value

$\rightarrow \cos\left(\frac{2k\pi + p\theta}{q}\right) + i\sin\left(\frac{2k\pi + p\theta}{q}\right)$

$n \in Q \rightarrow$ There will be q values

$\rightarrow \cos(2k\pi + \theta)n + i\sin(2k\pi + \theta)n$

$n \in Q^c \rightarrow$ there will be ∞ values

Ques: Find the roots of $x = (1)^{1/7}$

Solu: $x^7 - 1 = 0$

It should have 7 roots [Be it real or complex]

$x = (1)^{1/7}$

can be written as

$z = r(\cos\theta + i\sin\theta)^{1/7}$

Where $\theta = \tan^{-1}\left(\frac{0}{1}\right) = 0$

$r = 1$

$1/7$ is rational $\Rightarrow$

$$(\cos\theta + i\sin\theta)^{1/7} = \cos\left(\frac{2k\pi + p\theta}{q}\right) + i\sin\left(\frac{2k\pi + p\theta}{q}\right)$$

$p = 1,\ q = 7,\ \theta = 0$

$K = 0$ 1st root $\rightarrow$ 1

$K = 1$ 2nd root $\rightarrow$ $\cos\frac{2\pi}{7} + i\sin\frac{2\pi}{7}$

$K = 2$ 3rd root $\longrightarrow$ $\cos\frac{4\pi}{7} + i\sin\frac{4\pi}{7}$

⋮

$K = 7$ 7th root $\longrightarrow$ $\cos\frac{12\pi}{7} + i\sin\frac{12\pi}{7}$

Ques: Find the root of $z^4 = 1 + \sqrt{3}i$

Solu: $r = 2,\quad \theta = \pi/3$

$$z^4 = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$z = 2^{1/4}\underbrace{\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{1/4}}_{\text{4 values}}$$

$$z = 2^{1/4}\left[\cos\left(\frac{2k\pi + \pi/3}{4}\right) + i\sin\left(\frac{2k\pi + \pi/3}{4}\right)\right]$$

$K = 0$ 1st root $\rightarrow$ $z = 2^{1/4}\left[\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right]$

$K = 1$ 2nd root $\rightarrow$ $z = 2^{1/4}\left[\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right]$

K=2 Third root → $z = 2^{1/4}\left[\cos\frac{13\pi}{12} + i\sin\frac{13\pi}{12}\right]$

K=3 4th root → $z = 2^{1/4}\left[\cos\frac{19\pi}{12} + i\sin\frac{19\pi}{12}\right]$

Ques: Find z satisfying, $z^6 = \frac{1+5i}{3i-2}$

Solu: $z^6 = \frac{(1+5i)}{-2+3i} \times \frac{(-2-3i)}{(-2-3i)}$

$= \frac{(-2+15) + i(-3-10)}{4+9}$

$= \frac{13-13i}{13} = 1-i$

$z = (1-i)^{1/6}$

$z = \underbrace{\sqrt{2}^{1/6}\left(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4}\right)^{1/6}}_{\text{6 values}}$

$z = \sqrt{2}^{1/6}\left[\cos\left(\frac{2K\pi + 7\pi/4}{6}\right) + i\sin\left(\frac{2K\pi + 7\pi/4}{6}\right)\right]$

Roots → $z = \sqrt{2}^{1/6}\,\text{cis}\left(\frac{7\pi}{4}\right)$, $\sqrt{2}^{1/6}\,\text{cis}\left(\frac{15\pi}{24}\right)$,

$\sqrt{2}^{1/6}\,\text{cis}\left(\frac{23\pi}{24}\right)$, $\sqrt{2}^{1/6}\,\text{cis}\left(\frac{31\pi}{24}\right)$, $\sqrt{2}^{1/6}\,\text{cis}\left(\frac{39\pi}{24}\right)$

$\sqrt{2}^{1/6}\,\text{cis}\left(\frac{47\pi}{24}\right)$

CUBE ROOT OF UNITY

$z^3 = 1 \Rightarrow z = (1)^{1/3}$ $\quad r = 1, \theta = 0$

$z = \cos\frac{2K\pi}{3} + i\sin\frac{2K\pi}{3}$

Roots → $1, \quad \frac{-1+\sqrt{3}i}{2} \hookrightarrow w \quad , \quad \frac{-1-\sqrt{3}i}{2} \hookrightarrow w^2$

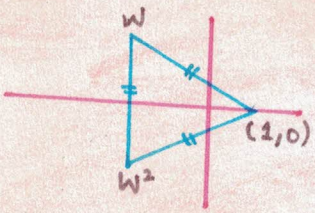
$z^3 - 1 = 0 \Leftarrow \begin{matrix} 1 \\ w \\ w^2 \end{matrix} \Rightarrow$

$1 + w + w^2 = 0$

$w + w^2 + w^3 = 0$

$w^3 = 1$

NOTES



i) 1, w, and w^2 form an eq. Δ

ii) $1 + x + x^2 = (x-w)(x-w^2)$

$1 - x + x^2 = (x+w)(x+w^2)$

iii) The roots of cube root of unity are diffred by an angle $2\pi/3$

iv) $1 + w^r + w^{2r} = \begin{cases} 3 & \text{If } r \text{ is multiple of } 3 \\ 0 & \text{otherwise} \end{cases}$

v) $\bar{w} = w^2$

vi) $w^{3k+1} = w \quad , \quad w^{3k+2} = w^2 \qquad [k \in I]$

vii) $a^3 - b^3 = (a-b)(a-bw)(a-bw^2)$

viii) $a^3 + b^3 = (a+b)(a+bw)(a+bw^2)$

ix) $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a+bw+cw^2)(a+bw^2+cw)$

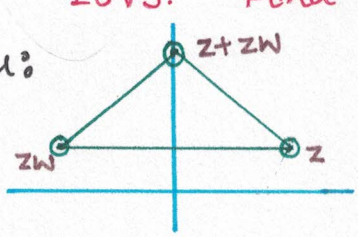
if a, b, c are sides of Δ, then

$a + bw + cw^2 = 0$

⇒ the Δ is equilateral Δ

Ques: Area of Δ formula by z, zw and z + zw is $16\sqrt{3}$. Find |z| ?

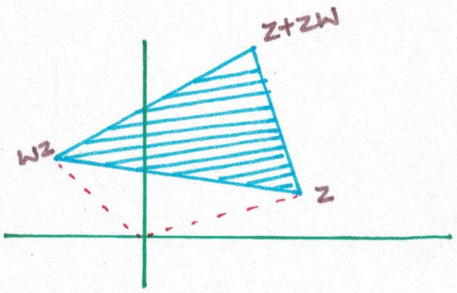
Soln:



$16\sqrt{3} = \frac{\sqrt{3}}{4} (\text{side})^2$

$16 \times 4 = (\text{side})^2$

$\text{side} = 8$



$$w = \frac{-1+\sqrt{3}i}{2}$$

$$z = \left(x-\frac{1}{2}\right) + i\left(y+\frac{\sqrt{3}}{2}\right)$$

∵ The Δ is a equilateral, sides are $z, zw, z(1+w)$ and

$$z+zw+z(1+w) = z(1+w+w^2) = 0$$

$$\boxed{|z| = 8}$$

Ques:- If $(a+w)^{-1} + (b+w)^{-1} + (c+w)^{-1} + (d+w)^{-1} = 2w^{-1}$

and $(a+w^2)^{-1} + (b+w^2)^{-1} + (c+w^2)^{-1} + (d+w^2)^{-1} = 2w^{-2}$

Then prove that: (i) $\sum abc = 2$

(ii) $\sum a = 2\pi a$

(iii) $(a+1)^{-1} + (b+1)^{-1} + (c+1)^{-1} + (d+1)^{-1} = 2$

Solu:- (i) $abc + bcd + cda + dab = 2$

Equation – $\frac{1}{a+x} + \frac{1}{b+x} + \frac{1}{c+x} + \frac{1}{d+x} - \frac{2}{x} = 0$

A biquadratic equation with roots w and w^2

$$(b+x)(c+x)(d+x)(x) + (a+x)(c+x)(d+x)x +$$
$$(a+x)(b+x)(d+x)x + (a+x)(b+x)(c+x)x$$
$$- 2(a+x)(b+x)(c+x)(d+x) = 0$$

METHOD OF FINDING THE SQUARE ROOT OF A COMPLEX NO.

$$\sqrt{a+ib} = x+iy$$

$$a+ib = x^2 - y^2 + 2ixy$$

$$x^2 - y^2 = a \quad , \quad 2xy = b$$

$$(x^2+y^2)^2 = a^2+b^2$$

$$x^2+y^2 = \sqrt{a^2+b^2}$$

Direct formula for square root of a complex no.

$$(x+iy)^{1/2} = \pm\left(\sqrt{\frac{|z| + Re(z)}{2}} \pm \sqrt{\frac{|z| - Re(z)}{2}}\right)$$

If $Im(z) > 0$, use (+) in between, if $Im(z) < 0$, use (−)

Ques: Find the square root of $-7-24i$

Solu: $(x^2+y^2) = 49 + 576 = 625$

$x^2+y^2 = 25 \qquad x^2-y^2 = -7$

$2x^2 = 18 \qquad 2xy = -24$

$x = \pm 3 \qquad y = \pm 4$

$\sqrt{z} = 3-4i \quad or \quad -3+4i$

Ques: $9+40i$

Solu: $81 + 1600 = 1681$

$x^2 + y^2 = 41$

$x^2 - y^2 = 9$

$2x^2 = 50$

$x = \pm 5$

$x^2 - y^2 = 9$

$25 - 9 = y^2$

$y = \pm 4$



Root → $5+4i, \; -5-4i$

Ques: $(-81)^{1/4}$

Solu: $z = (-81)^{1/4} \Rightarrow z^2 = (-81)^{1/2}$

$-81 = x+iy$

$z^2 = (+9i)^{2\times 1/2}$

$z^2 = 9i$

$z = \sqrt{9i}$

$x^2 - y^2 = 0$

$x^2 + y^2 = 9$

$x = \pm 3/\sqrt{2}$

$x^2 - y^2 = -81$

$2xy = 0$

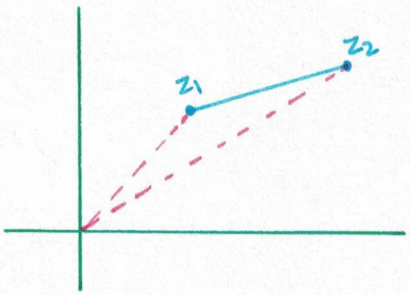
$x^2 = -81$

Root → $\frac{3}{\sqrt{2}}(1 \pm i)$

SIMILARITIES OF COMPLEX NUMBERS WITH CORDINATE GEOMATRY

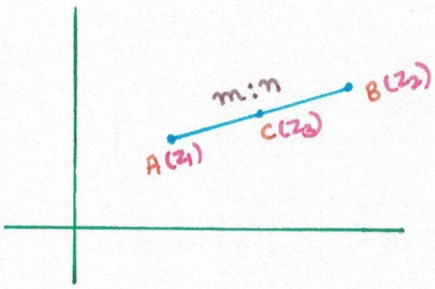
Complex numbers show similarity with 2D. It holds all the formulas of coordinate in complex form.

1

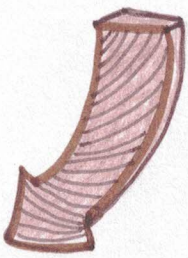


Distance Formula :- $|z_1 - z_2|$

2



Section Formula :- $z_3 = \frac{mz_2 + nz_1}{m+n}$



PROOF

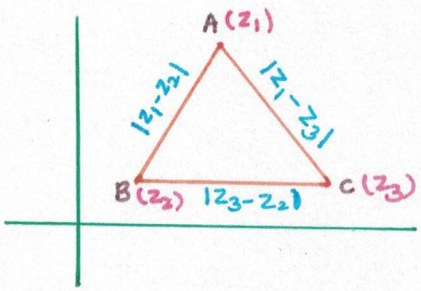
Rotate CB at an angle of π :

$$z_1 - z_3 = \frac{m(z_2 - z_3)}{n} e^{i\pi}$$

$$\Rightarrow z_1 n - z_3 n = (mz_2 - mz_3)(-1)$$

$$\Rightarrow z_3 = \frac{mz_2 + nz_1}{m+n}$$

3

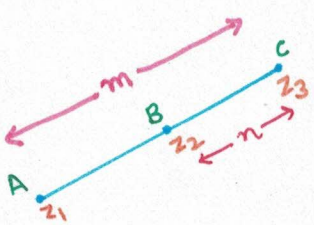


Centroid :- $\frac{z_1 + z_2 + z_3}{3}$

Incentre :-

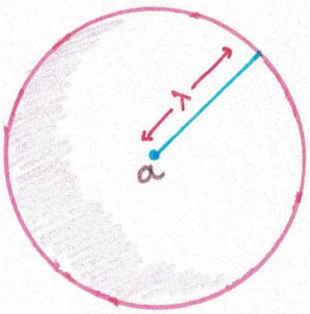
$$\frac{|z_2 - z_3| z_1 + |z_1 - z_3| z_2 + |z_1 - z_2| z_3}{|z_2 - z_3| + |z_1 - z_3| + |z_1 - z_2|}$$

4



External Division :- $z_3 = \frac{mz_2 - nz_1}{m-n}$

5 $|z-a| = \lambda$ is the equation of a circle where 'a' is center of the circle and 'λ' is the radius.

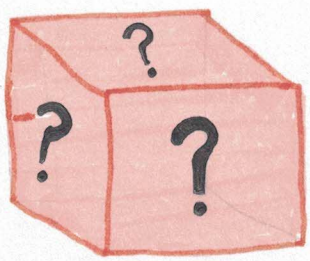


$$|z-a|^2 = \lambda^2$$

$$(z-a)(\overline{z-a}) = \lambda^2 \text{ or } (z-a)(\bar{z}-\bar{a}) = \lambda^2$$

$$\underbrace{z\bar{z}}_{x^2+y^2} - \underbrace{\bar{a}z - a\bar{z}}_{2gx+2fy} + \underbrace{a\bar{a}}_{c} = \lambda^2$$

Center of Circle = - Coeff. of $\bar{z}$



Ques: Find the equation of circle whose center is at $(2,-3)$ and radius is 3.

Sol: $[(x-2)+i(y+3)]^2 = 9$

$\Rightarrow x^2+y^2+4+9-4x+6y = 9$

$\Rightarrow x^2+y^2-4x+6y+4 = 0$

$(a = 2-3i)$ ↳

$\Rightarrow \boxed{z\bar{z} + (2+3i)z - (2-3i)\bar{z} + 4 = 0}$

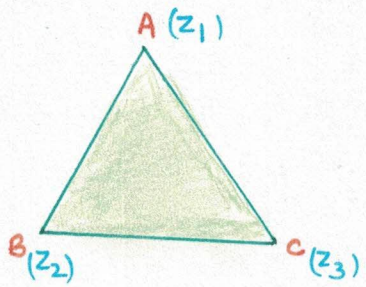
Ques: Find center and radius of circle

$$z\bar{z} - (1+i)z - (1-i)\bar{z} + 9 = 0$$

Sol: $C = (1,-1)$ $\quad 9 = 2-\lambda^2 \Rightarrow \lambda^2 = -7$

☞ This is not equation of a circle. Radius is not real.

6 Area of the triangle whose vertices are z_1, z_2 and z_3:-



$$\Delta = \left| \frac{i}{4} \begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix} \right|$$

mod — determinant

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$C_1 \longrightarrow C_1 + iC_2$

$$\Delta = \frac{1}{2}\begin{vmatrix} z_1 & y_1 & 1 \\ z_2 & y_2 & 1 \\ z_3 & y_3 & 1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} z_1 & z_1 - \bar{z}_1/z_i & 1 \\ z_2 & z_2 - \bar{z}_2/z_i & 1 \\ z_3 & z_3 - \bar{z}_3/z_i & 1 \end{vmatrix}$$

$$\Delta = \frac{1}{4i}\begin{vmatrix} z_1 & z_1 - \bar{z}_1 & 1 \\ z_2 & z_2 - \bar{z}_2 & 1 \\ z_3 & z_3 - \bar{z}_3 & 1 \end{vmatrix}$$

$$= \frac{-i}{4}\left\{ \begin{vmatrix} z_1 & z_1 & 1 \\ z_2 & z_2 & 1 \\ z_3 & z_3 & 1 \end{vmatrix} - \begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix} \right\}$$

$$\Delta = \left|\frac{i}{4}\begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix}\right|$$

Ques:- Find the area of Δ formed by $(1+i), (1+3i), (2+5i)$

Solu:-

$$\Delta = \frac{i}{4}\begin{vmatrix} 1+i & 1-i & 1 \\ 1+3i & 1-3i & 1 \\ 2+5i & 2-5i & 1 \end{vmatrix}$$

$$\Rightarrow \Delta = \frac{i}{4}\left[(1+i)(-1+2i) + (i-1)(-1-2i) + 1(2+15+i(-5+6)) - (2+15+i(5-6))\right]$$

$$\Rightarrow \Delta = \frac{i}{4}\left[-1-2+i(2-1)+1+2+i(1)+17-17+i+i\right]$$

$$\Rightarrow \Delta = \left|\frac{i}{4} \times 4i\right| = |i^2|$$

$\Delta = 1$

POINTS TO REMEMBER

If the area of Δ formed by z_1, z_2, z_3 is 0, then all 3 complex numbers are collinear, i.e. they lie on the

Same line. Hence we can find the equation of a line passing through z_1 & z_2

$$\begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \quad z(\bar{z}_1 - \bar{z}_2) + \bar{z}(z_2 - z_1) + 1(z_1\bar{z}_2 - z_2\bar{z}_1) = 0$$

$$\Rightarrow \quad \bar{A}z + A\bar{z} + C = 0$$

Its like,

Here are the complex form is given as above and the complex slope of straight line will be -

$$-\text{coeff of } \left(\frac{\bar{z}}{z}\right)$$

i.e. $$m = -\left(\frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2}\right)$$

There will be infinite points lying on a complex straight line, say $z_1, z_2, z_3 \ldots\ldots$, then -

$$-\left(\frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2}\right) = -\left(\frac{z_2 - z_3}{\bar{z}_2 - \bar{z}_3}\right) = \ldots\ldots\ldots\ldots m$$

If 2 complex lines are parallel then -

$$m_1 = m_2$$

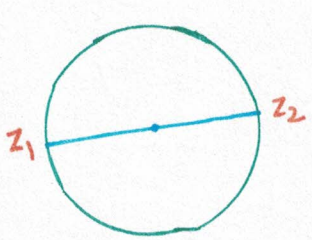
If 2 complex lines are perpendicular, then -

$$\frac{m_1}{m_2} = -1 \quad \text{or} \quad m_1 + m_2 = 0$$



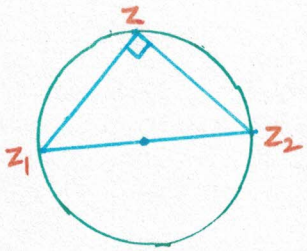
Ques: Find the equation of circle whose end points of diameter are z_1 & z_2

Solu: Radius $= \dfrac{|z_1 - z_2|}{2}$



$$\left|z - \left(\frac{z_1 + z_2}{2}\right)\right| = \frac{|z_1 - z_2|}{2}$$

$$\frac{z - z_1}{\bar{z} - \bar{z}_1} + \frac{z - z_2}{\bar{z} - \bar{z}_2} = 0$$



$$(z-z_1)(\bar{z}-\bar{z}_2) + (z-z_2)(\bar{z}-\bar{z}_1) = 0$$

Ques: Find the equation of a circle whose end points of the diameter are $3+4i$ & $1-i$

Solution: $u_1 + u_2 = 0$

$$\frac{z-(3+4i)}{\bar{z}-(\overline{3+4i})} + \frac{z-(1-i)}{z-(\overline{1-i})} = 0$$

Equation of tangent to circle $|z-a| = \lambda$ at α :-

Let α be any point on this circle, then equation of tangent is -

$$\frac{z-\alpha}{\bar{z}-\bar{\alpha}} + \frac{a-\alpha}{\bar{a}-\bar{\alpha}} = 0$$

$\Rightarrow (z-\alpha)(\bar{a}-\bar{\alpha}) + (\bar{z}-\bar{\alpha})(a-\alpha) = 0$

$\Rightarrow z\bar{a} - z\bar{\alpha} - \alpha\bar{a} + \alpha\bar{\alpha} + \bar{z}a - \bar{z}\alpha - \bar{\alpha}a + \alpha\bar{\alpha} = 0$

$\Rightarrow z(\bar{a}-\bar{\alpha}) + \bar{z}(a-\alpha) - \alpha\bar{a} - \bar{\alpha}a + 2\alpha\bar{\alpha} = 0$

$\Rightarrow z(\bar{\alpha}-\bar{a}) + \bar{z}(\alpha-a) + \alpha\bar{a} + \bar{\alpha}a - 2\alpha\bar{\alpha} = 0$

Equation of circle

$z\bar{z} - a\bar{z} - \bar{a}z + a\bar{a} - \lambda^2 = 0$

$\therefore \alpha$ lies on circle

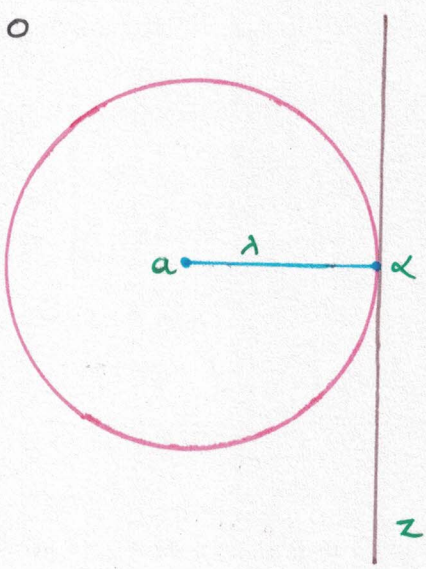
$z\bar{z} - a\bar{z} - \bar{a}z + a\bar{a} - \lambda^2 = 0$

$\Rightarrow \alpha\bar{\alpha} - a\bar{\alpha} - \bar{a}\alpha + a\bar{a} - \lambda^2 = 0$

$\Rightarrow \alpha\bar{\alpha} = a\bar{\alpha} + \bar{a}\alpha - a\bar{a} + \lambda^2$

$\Rightarrow z(\bar{\alpha}-\bar{a}) + \bar{z}(\alpha-a) + \alpha\bar{a} + \bar{\alpha}a - 2(\bar{\alpha}a + \bar{a}\alpha - a\bar{a} + \lambda^2) = 0$

$\Rightarrow \boxed{z(\bar{\alpha}-\bar{a}) + \bar{z}(\alpha-a) - \alpha\bar{a} - \bar{\alpha}a + 2a\bar{a} - 2\lambda^2 = 0}$



Trick: $z\bar{z} - a\bar{z} - \bar{a}z - \lambda^2 + a\bar{a} = 0$

$\Rightarrow 2z\bar{z} - 2a\bar{z} - 2\bar{a}z - 2\lambda^2 = 0$

$\Rightarrow (z\bar{z} + z\bar{z}) - (a\bar{z} + a\bar{z}) - (\bar{a}z + \bar{a}z) + 2a\bar{a} - 2\lambda^2 = 0$

$\Rightarrow (z\bar{\alpha} + \bar{z}\alpha) - (a\bar{\alpha} + a\bar{z}) - (\bar{a}\alpha + \bar{a}z) + 2a\bar{a} - 2\lambda^2 = 0$

$z(\bar{\alpha} - \bar{a}) + \bar{z}(\alpha - a) - a\bar{\alpha} - \bar{a}\alpha + 2a\bar{a} - 2\lambda^2 = 0$

Ques: Find the equation of tangent to the circle $|z| = 4$ at pt. α

Solution: $|z| = 4$

$z\bar{z} = 16$

$2z\bar{z} = 32$

$2z\bar{z} - 32 = 0$

$\boxed{z\bar{\alpha} + \alpha\bar{z} - 32 = 0}$

Ques:- Find the locus of z satisfying the equation $z^2 + (\bar{z})^2 = 2$

Solution: $z^2 + (\bar{z})^2 = 2 \Rightarrow x^2 - y^2 - 2ixy + x^2 - y^2 - 2ixy = 2$

$2x^2 - 2y^2 = 2$

$\boxed{x^2 - y^2 = 1}$

1 POINT OF INTERSECTION OF 2 COMPLEX LINES:

Let there be 2 equation of line in complex will be given as:

$$C \times (A\bar{z} + \bar{A}z + B) = 0$$
$$A \times (C\bar{z} + \bar{C}z + D) = 0$$

$$z(C\bar{A} + A\bar{C}) + BC - AD = 0$$

$$\boxed{z = \frac{AD - BC}{C\bar{A} - A\bar{C}}}$$

Ques: Find the point of intersection $(4+3i), (4+9i), (3+4i), (2+9i)$

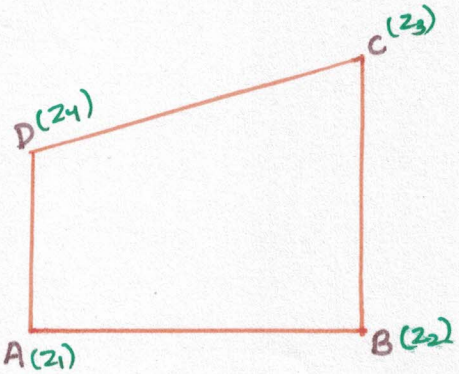
Solu: $z(\bar{z}_1 - \bar{z}_2) + \bar{z}(z_2 - z_1) + z_1\bar{z}_2 - z_2\bar{z} = 0$

$$\boxed{2(4 - 3i - (4 - 9i)) + \bar{z}(4 + 9i - 4 - 3i) + (4 + 3i)(4 - 9i) - (4 + 9i)(4 - 3i) = 0}$$

2 z_1, z_2, z_3, z_4 are cocyclic

$$\arg\left(\frac{z_4-z_1}{z_2-z_1}\right) + \arg\left(\frac{z_2-z_3}{z_4-z_3}\right) = \pi$$

$$\arg\left(\frac{z_4-z_1}{z_2-z_1} \times \frac{z_2-z_3}{z_4-z_3}\right) = \pi$$



$$\left(\frac{z_4-z_1}{z_2-z_1} \times \frac{z_2-z_3}{z_4-z_3}\right) \rightarrow \text{Real No.}$$

3 PERPENDICULAR DISTANCE OF A POINT α TO A COMPLEX LINE $A\bar{z} + \bar{A}z + b = 0$ WILL BE GIVEN BY

$$\left|\frac{a\bar{\alpha} + \bar{a}\alpha + b}{2|a|}\right|$$

PROOF:

$$a\bar{z} + \bar{a}z + b = 0$$

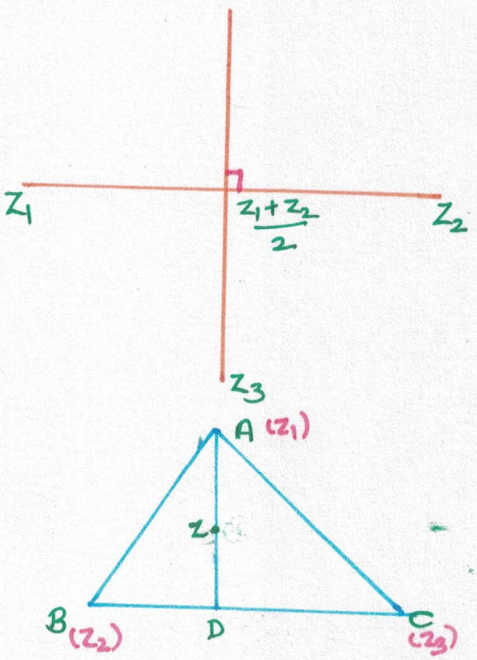
$$a = a_1 + ia_2 \quad , \quad z = x + iy \quad , \quad \alpha = \alpha_1 + i\alpha_2$$

$$P = \left|\frac{a\bar{\alpha} + \bar{a}\alpha + b}{2|a|}\right|$$

4 EQUATION OF PERPENDICULAR BISECTOR

If z_1 & z_2 are two end points of a complex line segment then the equation of perpendicular bisector will be given by-

$$\frac{z - \left(\frac{z_1+z_2}{2}\right)}{\bar{z} - \left(\frac{\bar{z}_1+\bar{z}_2}{2}\right)} + \frac{z_1-z_2}{\bar{z}_1-\bar{z}_2} = 0$$



5 ALTITUDES OF THE TRIANGLE

Equation of AD,

$$\frac{z-z_1}{\bar{z}-\bar{z}_1} + \frac{z_2-z_3}{\bar{z}_2-\bar{z}_3} = 0$$

6 ANGLE BISECTOR OF 2 COMPLEX NUMBER

Let the 2 lines be -

$A\bar{z} + \bar{A}z + B = 0$

$C\bar{z} + \bar{C}z + D = 0$

$$\frac{A\bar{z} + \bar{A}z + B}{2|A|} = \pm \frac{C\bar{z} + \bar{C}z + D}{2|C|}$$

EQUATION OF PARABOLA

Let z_f be the focus and $A\bar{z} + \bar{A}z + B = 0$ be the directrix, then the equation of parabola is :- $PS = PM$

$$|z - z_f| = \left| \frac{A\bar{z} + \bar{A}z + B}{2|A|} \right|$$

EQUATION OF ELLIPSE

Let z_f be the focus and $A\bar{z} + \bar{A}z + B = 0$ is a directrix & e is eccentricity then the equation of ellipse will be:-

$$|z - z_f| = e \left| \frac{A\bar{z} + \bar{A}z + B}{2|A|} \right| \qquad 0 < e < 1$$

EQUATION OF HYPERBOLA

$$|z - z_f| = e \left| \frac{A\bar{z} + \bar{A}z + B}{2|A|} \right| \qquad e > 1$$

Logarithm of complex number:

If we wish to find the logarithm of a complex no then,

$$\ln(x+iy) = a+ib$$

$$(x+iy) = e^{(a+ib)}$$

$$x+iy = e^a \cdot e^{ib}$$

$$x+iy = e^a(\text{cis } b)$$

$$x+iy = e^a(\cos b + i\sin b)$$

$$x = e^a \cos b \quad , \quad y = e^a \sin b$$

$$y/x = \tan b \quad \text{or} \quad b = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\ln(x^2+y^2) = 2a$$

$$a = \frac{1}{2}\ln(x^2+y^2)$$

$$\boxed{b = \arg(x+iy)}$$

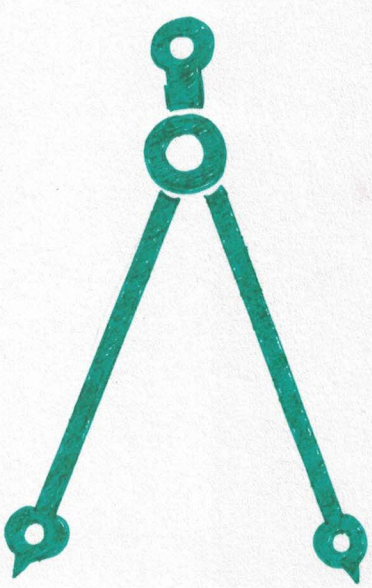
Ques: $y = i^i \qquad \ln y = i \ln i$

Sol: $\ln(0+i) = \frac{1}{2}\ln(1) + i\tan^{-1}\left(\frac{1}{0}\right)$

$$\ln i = i\,\pi/2$$

$$\ln y = -\pi/2$$

$$\boxed{y = e^{-\pi/2}}$$



Ques: $y = e^{e^{i\theta}}$

Sol: $\ln y = e^{i\theta}$

$$\ln y = \cos\theta + i\sin\theta$$

$$y = e^{\cos\theta + i\sin\theta}$$

$$\boxed{y = e^{\cos\theta} \cdot e^{i\sin\theta}}$$